M8-INMUX+

Improved Input Multiplexer for the Mark-8 Minicomputer

User's Guide

Rev A PC Boards

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Revision History

Manual	PCB	Date	Author	Notes

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OVERVIEW

The M8-INMUX+ is a replacement Input Multiplexer module for the Mark-8 minicomputer. It is electrically and mechanically compatible with the original Input Multiplexer designed by Jonathan Titus and may be installed in its place.

The M8-INMUX+ provides the following capabilities:

1. Support for all eight Intel 8008 input ports (INP 0 through INP 7). The original module implemented only two input ports.
2. Support for additional input ports on the Mark-8 bus, using the address lines, data lines, and the /IN strobe.
3. Two on-board input ports (INP 0 and INP 1), as on the original module. Each port may be enabled or disabled independently.
4. Drop-in replacement of the original board, with equivalent connectors and operating modes.

Theory of Operation and Design Philosophy

The original Input Multiplexer combined memory data, two on-board input ports, and the interrupt/JAM switch register using Signetics 8263 and 8267 multiplexers.

The 8263 is a 3-input, 4-bit multiplexer and can therefore select among only three sources (memory data and two input ports). The device is obsolete and is no longer practical for new construction.

The 8267 is a 2-input, 4-bit multiplexer (the open-collector counterpart of the 8266) used to select between the 8263 output and the interrupt/JAM port. It is likewise obsolete.

This design replaces those multiplexers with 74LS244 octal 3-state buffers. Each source—on-board INP 0, on-board INP 1, memory data, and JAM—is driven onto the CPU data bus by its own buffer. Output-enable logic ensures that only one source is active at a time.

Two on-board input ports (INP 0 and INP 1) are retained. Switch SW1 disables either port so that another card on the bus may occupy that input address.

Input Data on Bus

When this board replaces the original multiplexer, additional input ports may be implemented on the bus. An I/O card decodes A0–A2, A3 (DEN), and /IN, then drives D0–D7 with a 74LS244 or equivalent 3-state octal driver. The corresponding on-board port on this module must be disabled.

ASSEMBLY AND TEST INSTRUCTIONS

General Assembly Tips

Install components in order of increasing height, beginning with the lowest-profile parts. This allows the work surface to hold the parts against the board when the assembly is inverted for soldering.

Refer to the PC Board Drawings (page 11) for the location of each component.

Install all components of a given type before proceeding to the next type (for example, all resistors, then all IC sockets). Secure the parts with low-adhesion tape (painter's tape) before inverting the board for soldering.

When installing DIP sockets and DIP switches, seat each device fully against the board. Solder two diagonally opposite pins first. After all such devices are tacked in place, solder the remaining pins.

Assembly

Step 1 — Install resistors

Component	Value	Location	Quantity
Resistor, 1/4W	1K Ω	R1	1
Resistor, 1/4W	4.7K Ω	R2-R3	2

Step 2 — Install DIP sockets

Component	Location	Quantity
14-pin DIP socket, 0.3" wide	U5-U6	2
16-pin DIP socket, 0.3" wide	U7	1
20-pin DIP socket, 0.3" wide	U1-U4	4

See General Assembly Tips above. Do not install a socket for the DIP switch. Tack each socket with two diagonally opposite pins, and complete soldering after the DIP switch has been similarly tacked in place.

Step 3 — Install DIP switches

Component	Location	Quantity
DIP Switch, 2-Position SPST	SW1	1

Step 4 — Install ceramic capacitors

Component	Value	Location	Quantity
Capacitor, Ceramic	0.1 uF	C1-C7	7

Step 5 — Install electrolytic capacitors

Component	Value	Location	Quantity
Capacitor, electrolytic, 16 V, axial	10 uF	C8	1

Observe polarity. The positive lead is marked “+” on the PCB.

Step 6 — Install connectors

Seat each connector flush with the board. Solder one pin at each end, verify alignment, then solder the remaining pins.

J10 is not a standard single-piece 41-position connector. Assemble it from standard Molex 0.156 in housings; for example, three 10-position connectors and one 11-position connector.

Component	Location	Quantity
Right-angle female connector, Molex 0.156 in, 8-position	J1,J8,J11	3
Right-angle female connector, Molex 0.156 in, 2-position	J12	1
Right-angle female connector, Molex 0.156 in, 41-position	J10	1

Step 7 — Install jumpers

Rev A requires two jumper wires, as listed below.

From	To
U3 Pin 1	U5 Pin 6
U4 Pin 20	U7 Pin 16

Step 8 — Inspect the assembly

1. Inspect all solder joints for omissions and bridges.
2. With an ohmmeter, check for a short between J10 pin 3 (GND) and pin 6 (+5 V). Installed components produce a finite resistance; a near-zero reading indicates a short circuit, which must be corrected before proceeding.

Step 9 — Clean the board

If organic-core or water-soluble-flux solder was used, cleaning is required. To remove water-soluble flux:

1. Wash the board after soldering is complete and before ICs are inserted. Hold the board under running hot water and scrub with a brush for several minutes.
2. Remove water with compressed air, including from beneath components and from inside IC sockets. A workshop compressor is more effective than aerosol dusters.
3. Dry the board near a moderate heat source for approximately 30 minutes.

If rosin-core solder was used, cleaning is optional. Apply a flux remover suitable for rosin; a brush may be used to loosen residue. Inspect the solder joints again after cleaning.

Step 10 — Install ICs

Insert each IC with pin 1 toward the top of the board, as shown on the PC Board Drawings. Do not fold pins under the body. If necessary, press each row of pins against the work surface to set the lead pitch before insertion.

Refer to the PC Board Drawings for IC locations and orientation.

Component	Value	Location	Quantity
IC, DIP, Octal Buffer/Line Driver	74LS244	U1-U4	4
IC, DIP, Quad 2-Input NAND Gate	74LS00	U5	1
IC, DIP, Hex Inverter	74LS04	U6	1

IC, DIP, 3-to-8 decoder	74LS138	U7	1

Step 11 — Configuration

Configure SW1 as follows. A value of 0 denotes open (off); 1 denotes closed (on).

Switch	Setting	Switch Function
SW1<2:1>	11	Enables on-board INP 0 and INP 1

Setting a switch to 0 (open) disables the corresponding on-board port so that a bus card may use that INP address.

Step 12 — Final inspection

Inspect the completed board for the following defects:

- IC installed with reversed orientation
- IC pin folded under the package
- Incorrect device in any socket
- Incorrect switch configuration

Testing

The following procedure verifies basic operation of the M8-INMUX+ in a working Mark-8 system. It is not a complete troubleshooting guide. Diagnosis of a non-functional system is beyond the scope of this manual.

Step 1 — Installation

Remove the original Input Multiplexer from the Mark-8 backplane and install the M8-INMUX+ in the same slot. Remove other non-essential I/O cards for this test.

Step 2 — Connections

Connect INTERRUPT INPUT, MEMORY DATA, and JAM as on the original multiplexer. JAM is the left pin of the 2-position Molex connector (J12).

Apply a known pattern to on-board input port 0 by tying the eight data bits alternately to +5 V and GND, for example 10101010 (252 octal). This will be used to test the input port.

Step 3 — Input-port test

Apply power and enter the following program at memory address 0:

Address	Machine Code	Assembly Code
0	101	INP 0
1	121	OUT 0
2	377	HLT

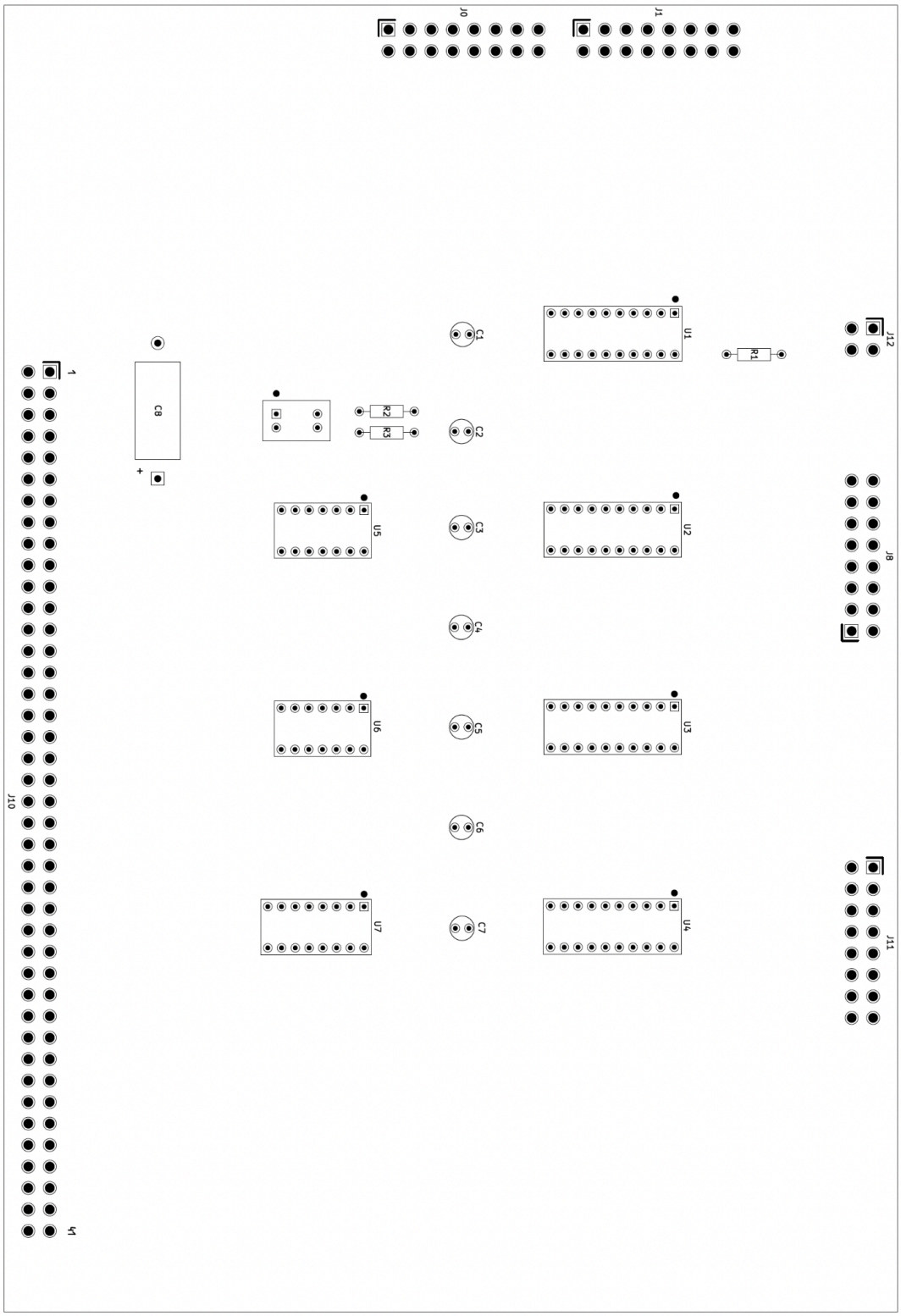
Execute from address 0 (interrupt with 105 octal). After the processor halts, output port 0 (last row of LEDs on the front-panel) should display 10101010 (252 octal).

PC BOARD COMPONENTS

Component	Value	Location	Quantity
PC Board, M8-INMUX+, Rev A			1
Resistor, 1/4W	1K Ω	R1	1
Resistor, 1/4W	4.7K Ω	R2-R3	2
14-pin DIP socket, 0.3" wide		U5-U6	2
16-pin DIP socket, 0.3" wide		U7	1
20-pin DIP socket, 0.3" wide		U1-U4	4
DIP Switch, 2-Position SPST		SW1	1
Capacitor, Ceramic	0.1 uF	C1-C7	7
Capacitor, electrolytic, 25 V, axial	100 uF	C8	1
Right-angle female connector, Molex 0.156 in, 8-position		J1,J8,J11	3
Right-angle female connector, Molex 0.156 in, 2-position		J12	1
Right-angle female connector, Molex 0.156 in, 41-position (three 10-position and one 11-position)		J10	1
IC, DIP, Octal Buffer/Line Driver	74LS244	U1-U4	4
IC, DIP, Quad 2-Input NAND Gate	74LS00	U5	1
IC, DIP, Hex Inverter	74LS04	U6	1
IC, DIP, 3-to-8 decoder	74LS138	U7	1

PC BOARD DRAWINGS

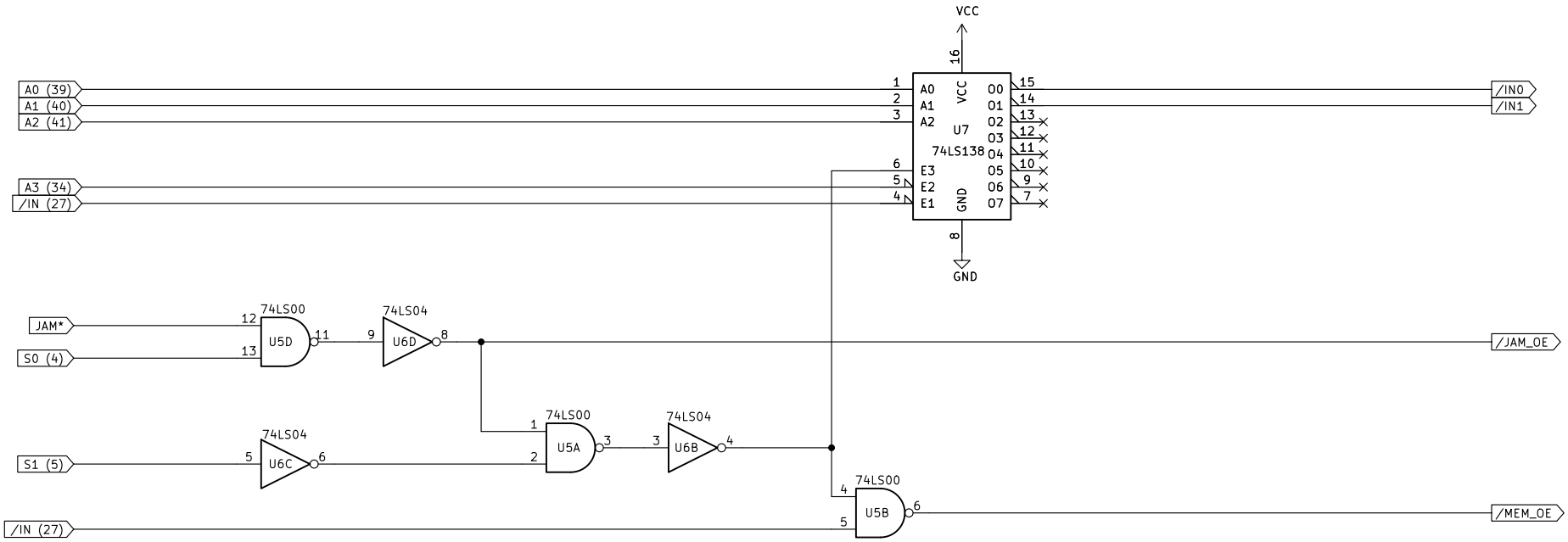
The following pages show component placement for Rev A.

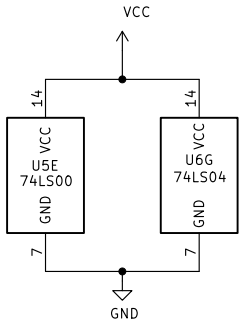


Rev A Component Placement

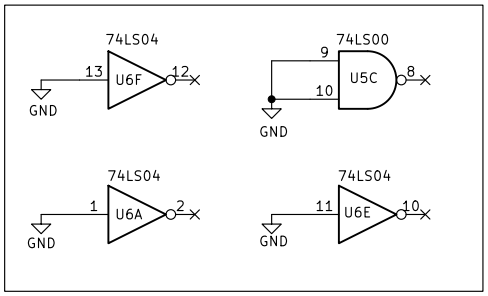
SCHEMATIC DRAWINGS

The following pages contain the schematic for Rev A.





SPARE GATES



Mark-8 Improved Input Multiplexer

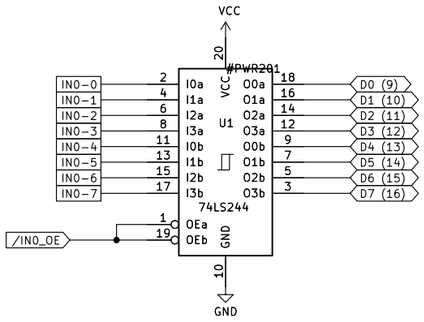
DESIGNED BY ARTEM KALINCHUK

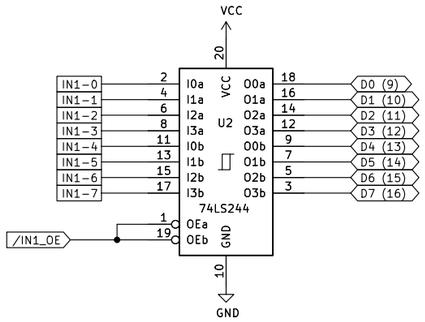
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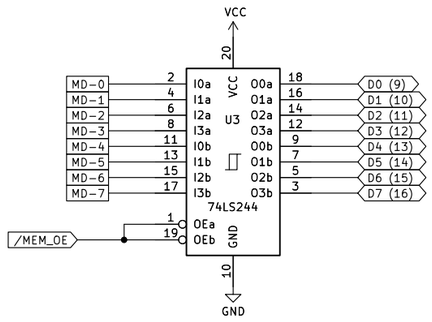
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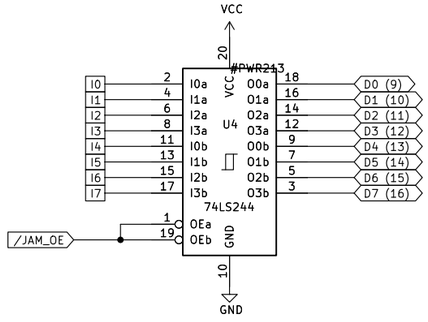
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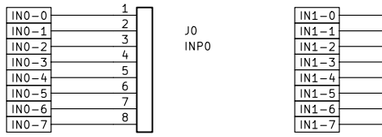
Mark-8 Improved Input Multiplexer

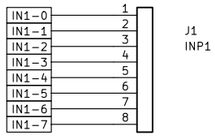
DESIGNED BY ARTEM KALINCHUK

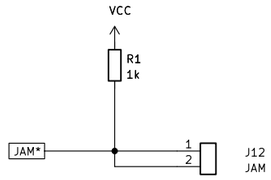
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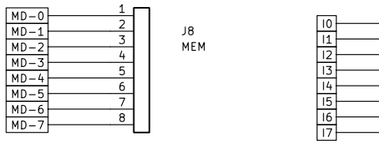
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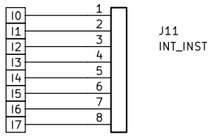
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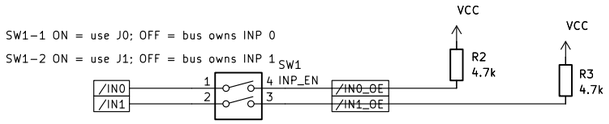


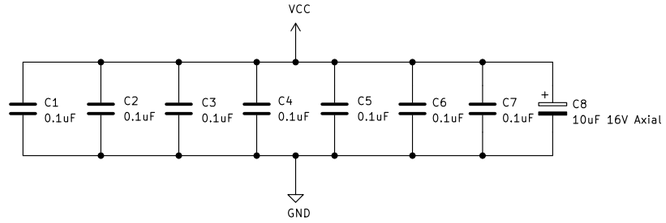


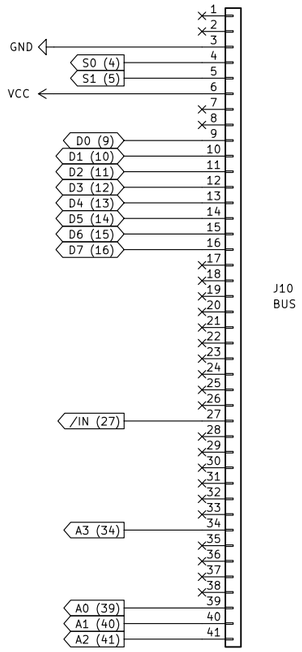












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Sheet: /Connectors / Power/
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